CellSense

An AI Assistant for Google Sheets

Aalto-yliopisto
Aalto-universitetet
Aalto University

Vu Minh Nguyen

24th February 2026

Introduction

Problem

- Google Sheets is a popular tools, so many people from different backgrounds use it.
- Therefore, people without proficiency in office software often struggles with remebering complex syntax for formulas.
- To address this, this project developed an AI-powered assistant to assist users in generating formulas.

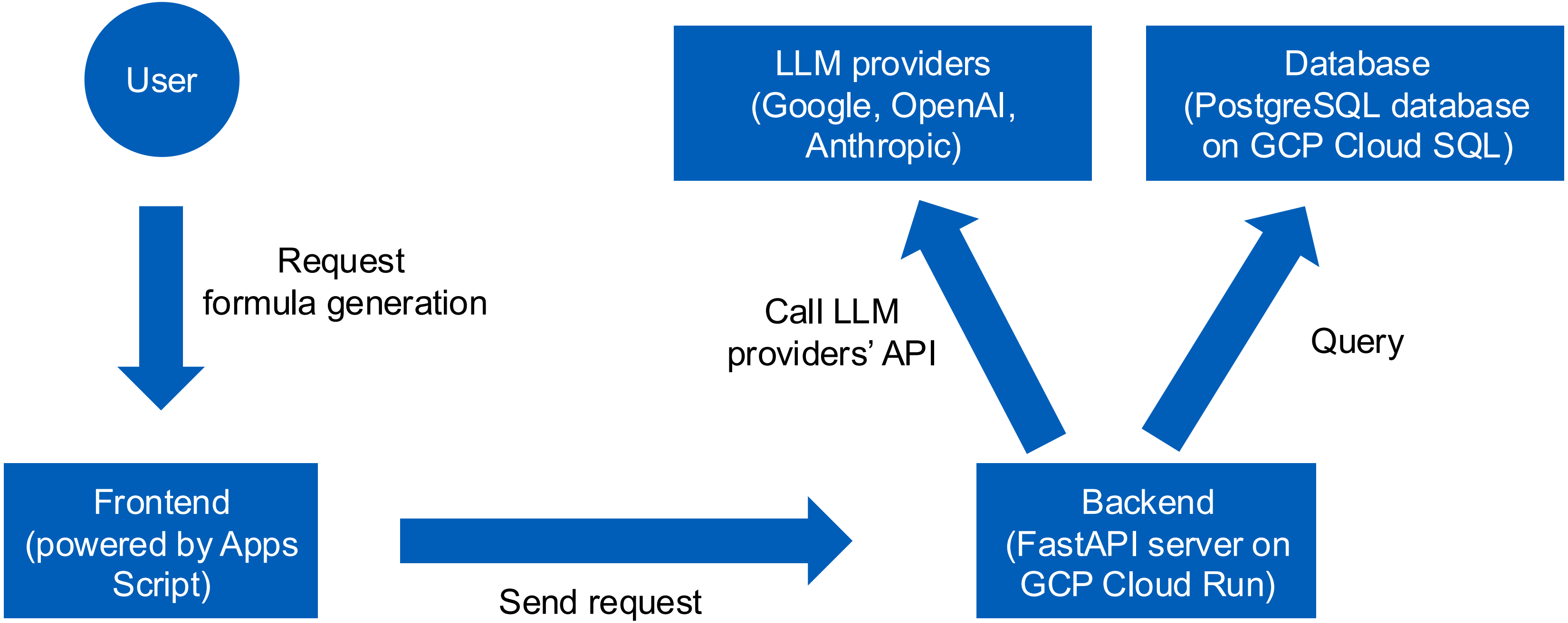
Objectives

The final version should be capable of:

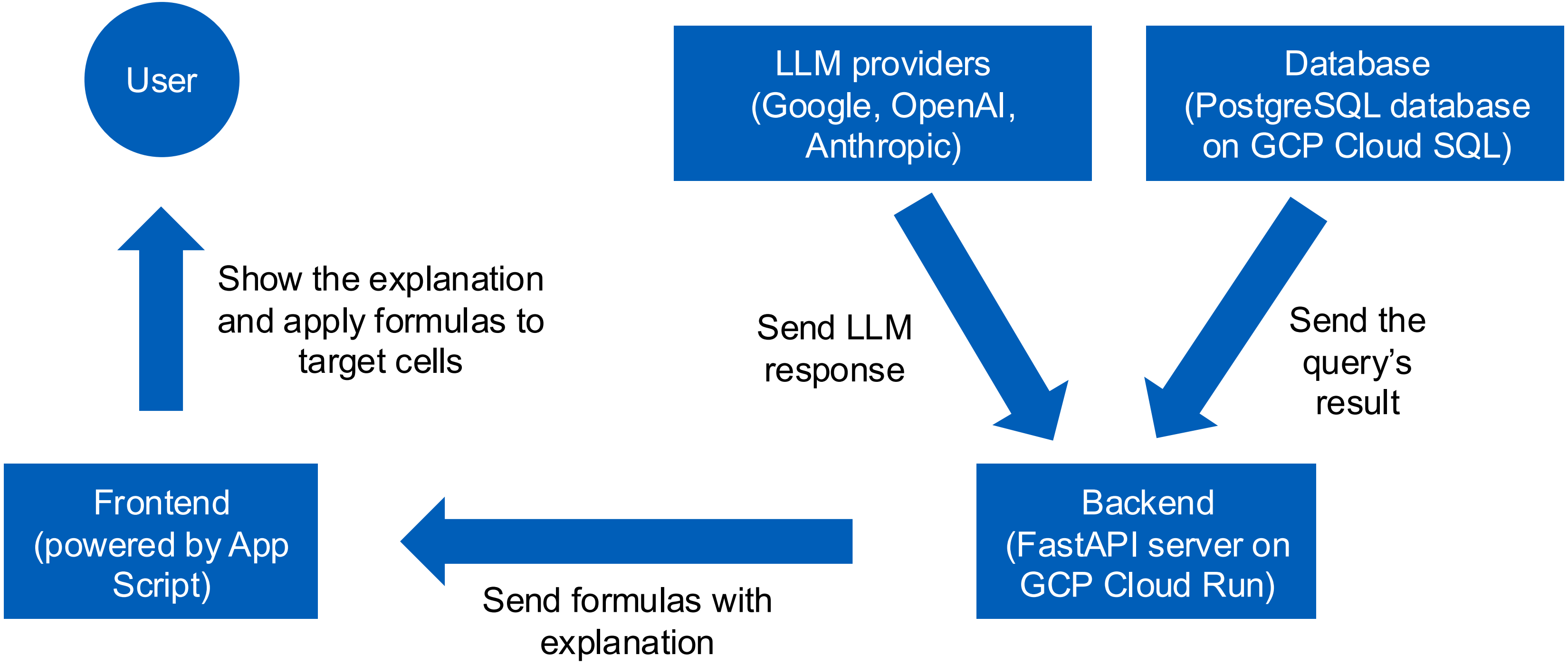
- Generating formulas according to the user's instructions. Attaching previous messages in the chat as context for multi-turn conversations.
- Supporting multiple LLM providers (Google Gemini, OpenAI GPT, Anthropic Claude).
- Allowing users to input their own API keys for their preferred LLM provider.
- And more… (in the final report)

System Architecture

Overview


User
Request
formula generation
Frontend
(powered by Apps
Script)
Send request
Backend
(FastAPI server on
GCP Cloud Run)
Call LLM
providers' API
LLM providers
(Google, OpenAI,
Anthropic)
Query
Database
(PostgreSQL database
on GCP Cloud SQL)

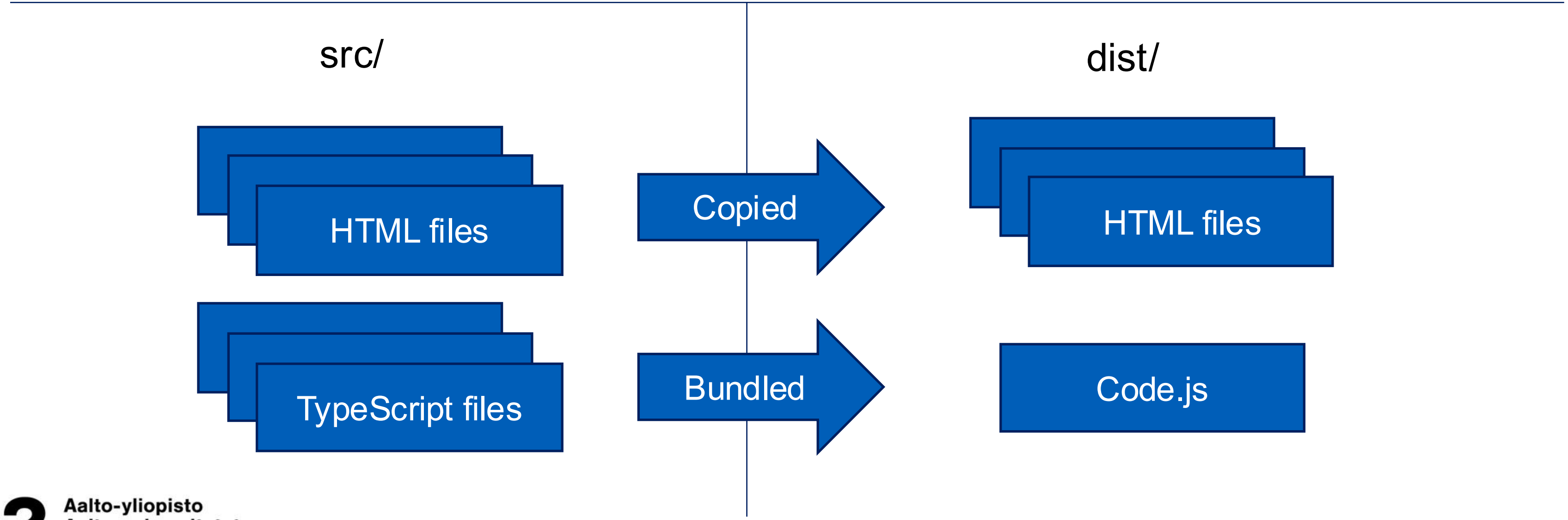
Overview



Aalto-yliopisto
Aalto-universitetet
Aalto University

Frontend

The frontend is a Google Apps Script add-on written in TypeScript, embedded directly into Google Sheets as a sidebar.



Backend

The frontend is FastAPI server hosted on GCP Cloud Run

App

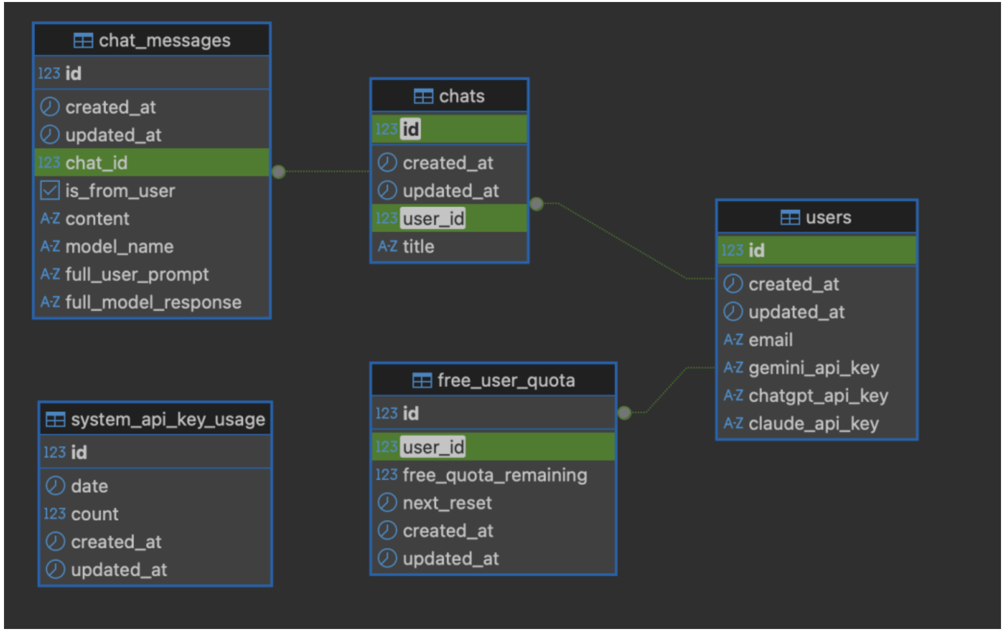
Router layer

Service layer — where most logics happen, may call LLM API

CRUD layer — queries database

Database

The databse is a Postgres database hosted on GCP Cloud SQL



Aalto-yliopisto
Aalto-universitetet
Aalto University

Cloud Infrastructure

- Google Cloud Platform (GCP) is used for hosting the backend and database.
- Resources:
  - **Google Cloud Run** is used to host the backend and to run jobs
  - **Google Container Registry** is used for storing Docker images
  - **Google Cloud SQL for PostgreSQL** is for database
  - **Google Apps Script** is used for the frontend, deployed as a Google Sheets add-on

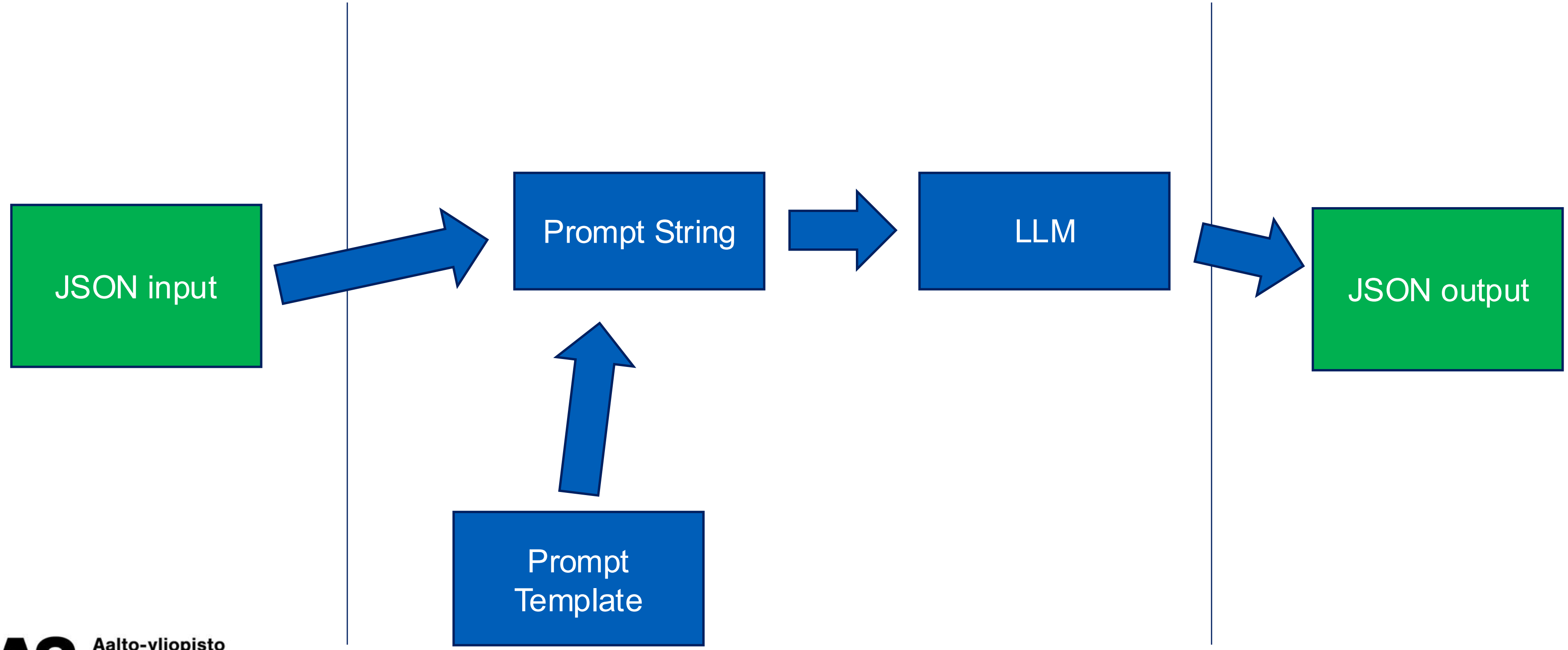
LLM Integration

Supported Models

- **Google:** gemini-2.5-pro, gemini-2.5-flash
- **OpenAI:** gpt-5, gpt-5-mini
- **Anthropic:** claude-opus-4-5, claude-haiku-4-5

During the LLM evaluation, some incapable models are excluded, including: gemini-2.5-flash-lite, gpt-5-nano, and claude-sonnet-4-5

Structured Input, Structured Output



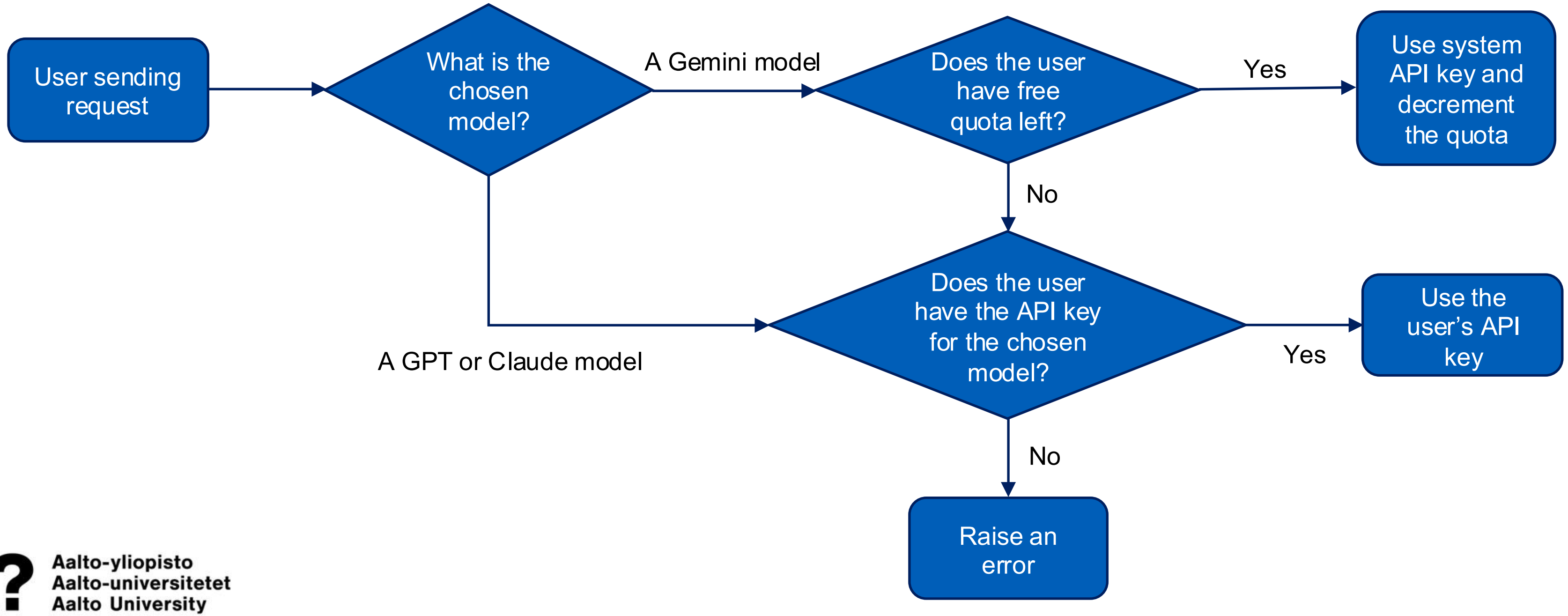
User and API Key Management

User Management

Users are automatically created on their first API request:

- Upon receiving a request, the system check if there is any user with the email in the request.
- If no user exists, one is created automatically.

API Key and Quota Management



Aalto-yliopisto
Aalto-universitetet
Aalto University

Security

OpenID Connect (OIDC)

- The frontend (Apps Script) attaches to all requests an OIDC token of the effective user (the logged in user).
- OIDC is a JSON web token (JWT) signed by Google.
- The backend then decode the JWT token and confirm the signature.
- **… at this point, we can confirm that the token has not been tampered, and it is indeed issued by Google**

OpenID Connect (OIDC)

- The body of the decoded token contain the effective user’s email, audience (target client ID), and other fields.
- The effective user’s email is used to create account (if not existed).
- The audience should be the OAuth ID of the Apps Script.
- **… at this point, we can confirm that the token is issued for the Google Apps Script of our app, and we can authenticate the logged in user**

Rejecting old request

- All requests older than 1 minutes are rejected. This prevent replay/delay attacks.

Aalto-yliopisto
Aalto-universitetet
Aalto University

Thank you!

aalto.fi